22AIE213

Machine Learning

HEART DISEASE PREDICTION USING MACHINE LEARNING

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Introduction

- Cardiovascular diseases are among the leading causes of death worldwide, highlighting the importance of early diagnosis and prevention.
- Analyzing medical factors using conventional approaches can be complex and time-consuming, particularly when dealing with large amounts of clinical data.
- Machine Learning techniques can analyze medical datasets efficiently and identify patterns associated with heart disease.
- In this project, machine learning algorithms are applied to a clinical heart attributes and develop a predictive system for heart disease detection.

Literature review

Paper Title	***Contribution***	***Research Gap***
Prediction of Heart Disease Using Machine Learning Techniques	Implemented several ML models to predict heart disease and compared their performance.	Limited focus on **data preprocessing and imbalance handling**.
Heart Disease Prediction Using Machine Learning Algorithms	Developed predictive models using algorithms such as Decision Tree and Random Forest.	Lack of **comprehensive model comparison and feature scaling techniques**.
Comparative Analysis of Machine Learning Algorithms for Heart Disease Prediction	Compared multiple classification models to evaluate prediction accuracy.	**Class imbalance and advanced preprocessing methods** were not sufficiently addressed.
An Efficient Heart Disease Prediction System Using Machine Learning	Proposed a machine learning based system for predicting heart disease risk.	Limited emphasis on **systematic exploratory analysis and robust evaluation metrics**.

Problem Statement

- Accurate prediction of heart disease requires the simultaneous analysis of several clinical and physiological indicators, which makes the diagnostic process complex.
- The relationships among these medical attributes are often non-linear and difficult to interpret using conventional analytical methods.
- Traditional approaches may not effectively utilize advanced computational techniques to identify hidden patterns and predictive relationships in medical data.
- Therefore, there is a need to develop a robust machine learning–based framework that can analyze clinical attributes and assist in reliable heart disease risk prediction.

Objectives

- To develop a machine learning-based system that can predict the risk of heart disease using clinical attributes.
- To ensure high-quality input data through EDA and preparation of clinical features.
- To address limitations observed in previous studies, particularly insufficient data preparation and imbalance handling.
- To evaluate and compare different machine learning models to determine the most effective model for prediction.

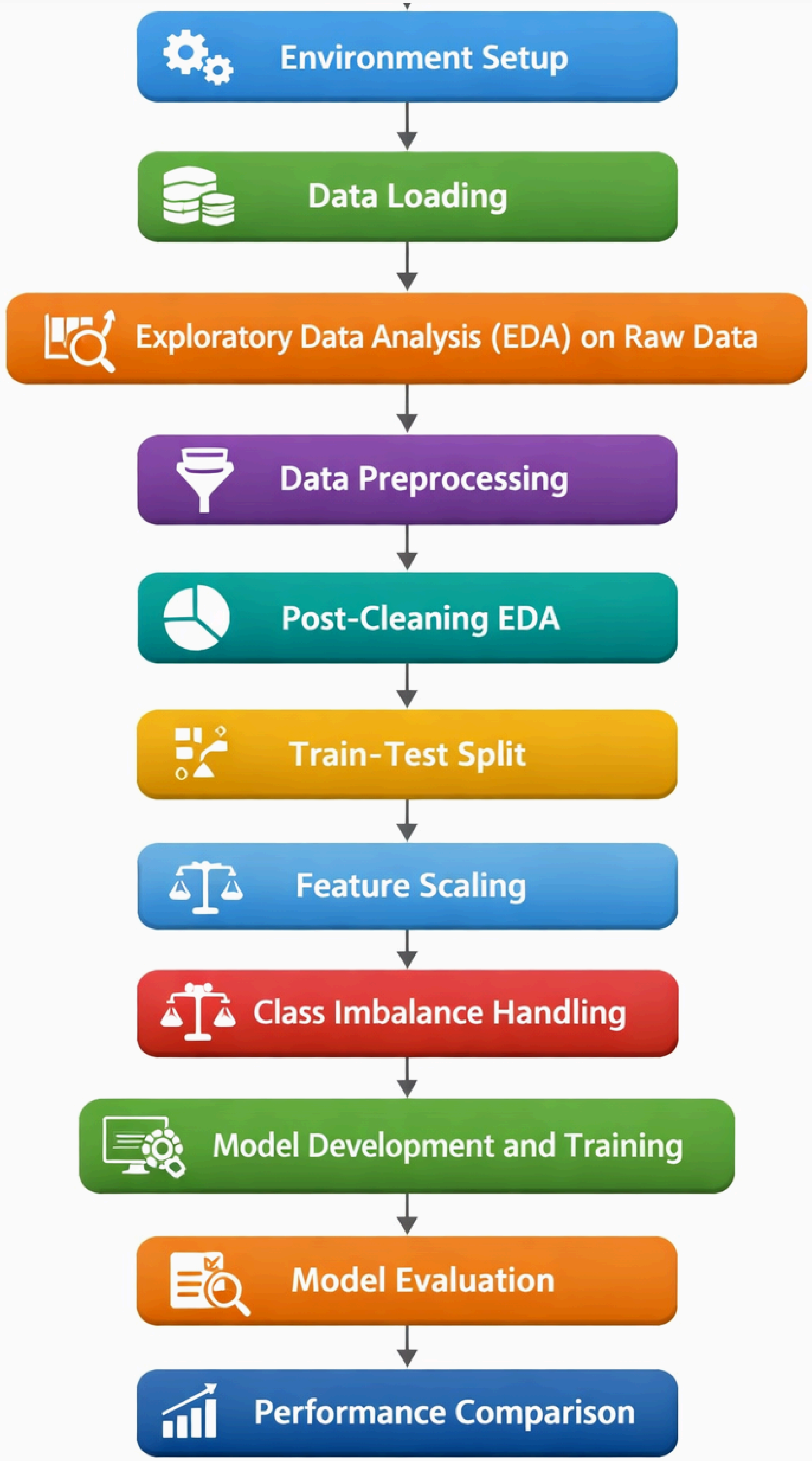
Dataset description

- The dataset contains 920 patient records with 16 attributes, representing clinical and physiological parameters related to heart health.
- Source - UCI Machine Learning Repository
- The attributes include demographic information, clinical measurements, and diagnostic test results collected from multiple medical centers.
- The variable ‘num’ represents the diagnosis of heart disease and serves as the target variable for prediction in this study.
- It contains five classes (0–4) indicating the severity level of heart disease.

  0 → No presence of heart disease

  1–4 → Presence of heart disease with increasing severity levels

Project Pipeline



Methodology

1.Environment Setup

Essential libraries were imported for data processing, visualization, machine learning, and model evaluation.

- Pandas – Used for data loading and tabular data manipulation.
- NumPy – Used for numerical computations and array operations.
- Matplotlib – Used for basic data visualization.
- Seaborn – Used for statistical data visualization.
- Scikit-learn – Used for preprocessing, model training, and evaluation.
- Imbalanced-learn (SMOTE) – Used to handle class imbalance.
- XGBoost – Used for gradient boosting–based predictive modeling.
- LightGBM – Used for efficient gradient boosting on structured data.

2. Data Loading

- The dataset was imported into the Python environment using Pandas data analysis library.

3.Exploratory Data Analysis (EDA) on Raw Data

- Analyzed the dataset dimensions to determine the number of observations and features i.e (920,16) using ‘df.shape’ .
- Examined the dataset structure, including feature names, data types, and non-null value counts using ‘df.info()’ .

```
RangeIndex: 920 entries, 0 to 919
Data columns (total 16 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   id        920 non-null    int64
 1   age       920 non-null    int64
 2   Gender    920 non-null    object
 3   dataset   920 non-null    object
 4   cp        920 non-null    object
 5   trestbps  861 non-null    float64
 6   chol      890 non-null    float64
 7   fbs       830 non-null    object
 8   restecg   918 non-null    object
 9   thalch    865 non-null    float64
 10  exang     865 non-null    object
 11  oldpeak   858 non-null    float64
 12  slope     611 non-null    object
 13  ca        309 non-null    float64
 14  thal      434 non-null    object
 15  num       920 non-null    int64
dtypes: float64(5), int64(3), object(8)
```

- Identified and separated categorical and numerical features based on their data types.

```
Categorical columns: ['Gender', 'dataset', 'cp', 'fbs', 'restecg', 'exang', 'slope', 'thal']
Numerical columns: ['id', 'age', 'trestbps', 'chol', 'thalch', 'oldpeak', 'ca', 'num']
```

- Generated descriptive statistics to summarize the distribution and variability of numerical features using ‘df.describe()’

	id	age	trestbps	chol	thalch	oldpeak	ca	num
count	920.000000	920.000000	861.000000	890.000000	865.000000	858.000000	309.000000	920.000000
mean	460.500000	53.510870	132.132404	199.130337	137.545665	0.878788	0.676375	0.995652
std	265.725422	9.424685	19.066070	110.780810	25.926276	1.091226	0.935653	1.142693
min	1.000000	28.000000	0.000000	0.000000	60.000000	-2.600000	0.000000	0.000000
25%	230.750000	47.000000	120.000000	175.000000	120.000000	0.000000	0.000000	0.000000
50%	460.500000	54.000000	130.000000	223.000000	140.000000	0.500000	0.000000	1.000000
75%	690.250000	60.000000	140.000000	268.000000	157.000000	1.500000	1.000000	2.000000
max	920.000000	77.000000	200.000000	603.000000	202.000000	6.200000	3.000000	4.000000

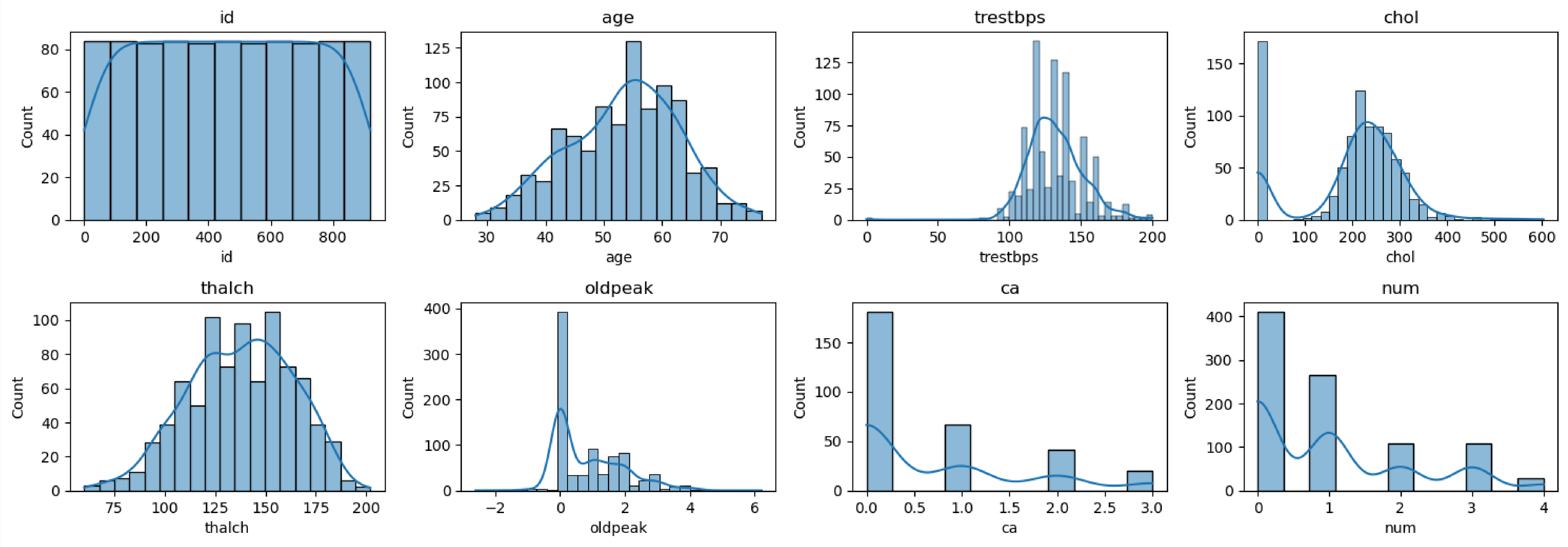
Analyzed the distribution of the target variable to determine the frequency of each class.

```
num
0    411
1    265
2    109
3    107
4     28
Name: count, dtype: int64
```

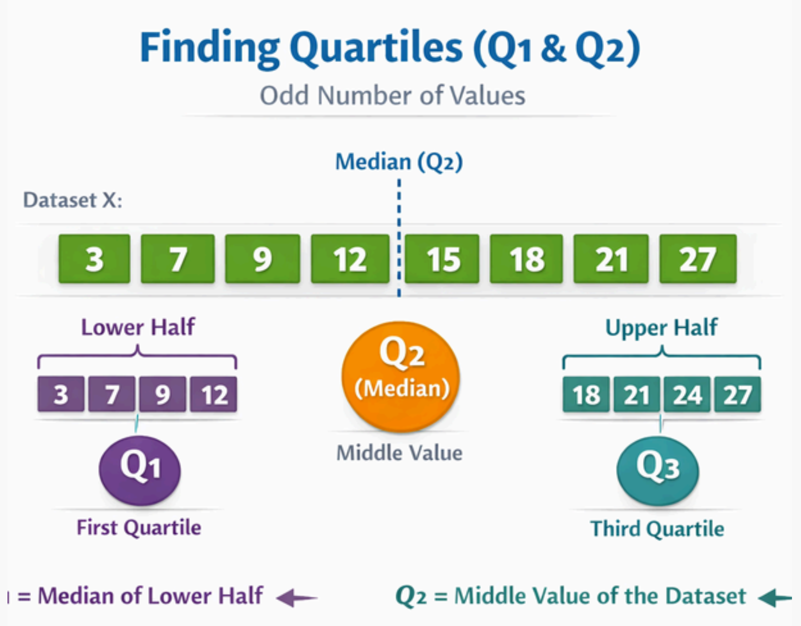
- Calculated and analyzed the percentage of missing values in each feature to assess data completeness.

	Missing Values	Percentage (%)
ca	611	66.413043
thal	486	52.826087
slope	309	33.586957
fbs	90	9.782609
oldpeak	62	6.739130
trestbps	59	6.413043
thalch	55	5.978261
exang	55	5.978261
chol	30	3.260870
restecg	2	0.217391
id	0	0.000000
age	0	0.000000
Gender	0	0.000000
dataset	0	0.000000
cp	0	0.000000
num	0	0.000000

- **Univariate Distribution Analysis of Numerical Features** - Visualized the distribution of numerical features using histograms to understand their spread and underlying patterns.



- **Outlier Analysis Using Boxplots** - Used boxplots to identify potential outliers in numerical features.

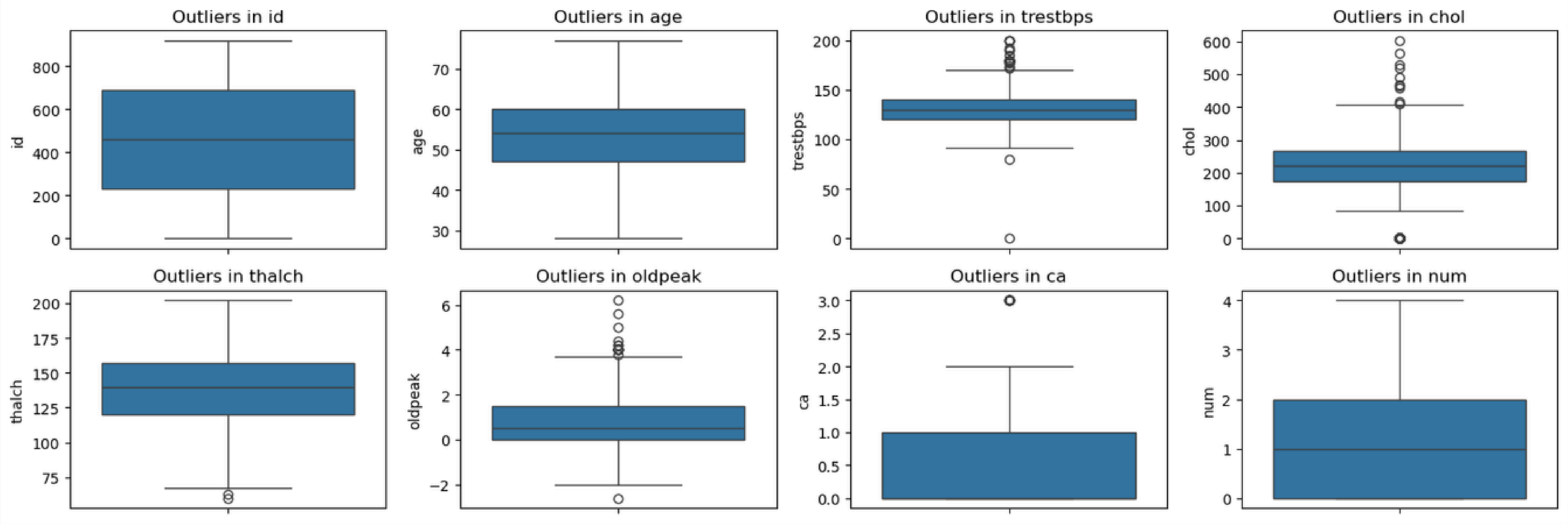


① Step 1: $IQR = Q_3 - Q_1$

② Step 2: Lower Bound $= Q_1 - 1.5\,(IQR)$
$= Q_1 - 1.5\,(Q_3 - Q_1)$
$= Q_1 - 1.5\,Q_3 + 1.5\,Q_1$
$= 2.5Q_1 - 1.5\,Q_3$

③ Step 3: Upper Bound $= Q_3 + 1.5\,(IQR)$
$= Q_3 + 1.5\,(Q_3 - Q_1)$
$= Q_3 + 1.5\,Q_3 - 1.5\,Q_1$
$= 2.5\,Q_3 - 1.5\,Q_1$

④ Step 4: Outlier Condition:
$x < Q_1 - 1.5\,(IQR)$
OR
$x > Q_3 + 1.5\,(IQR)$



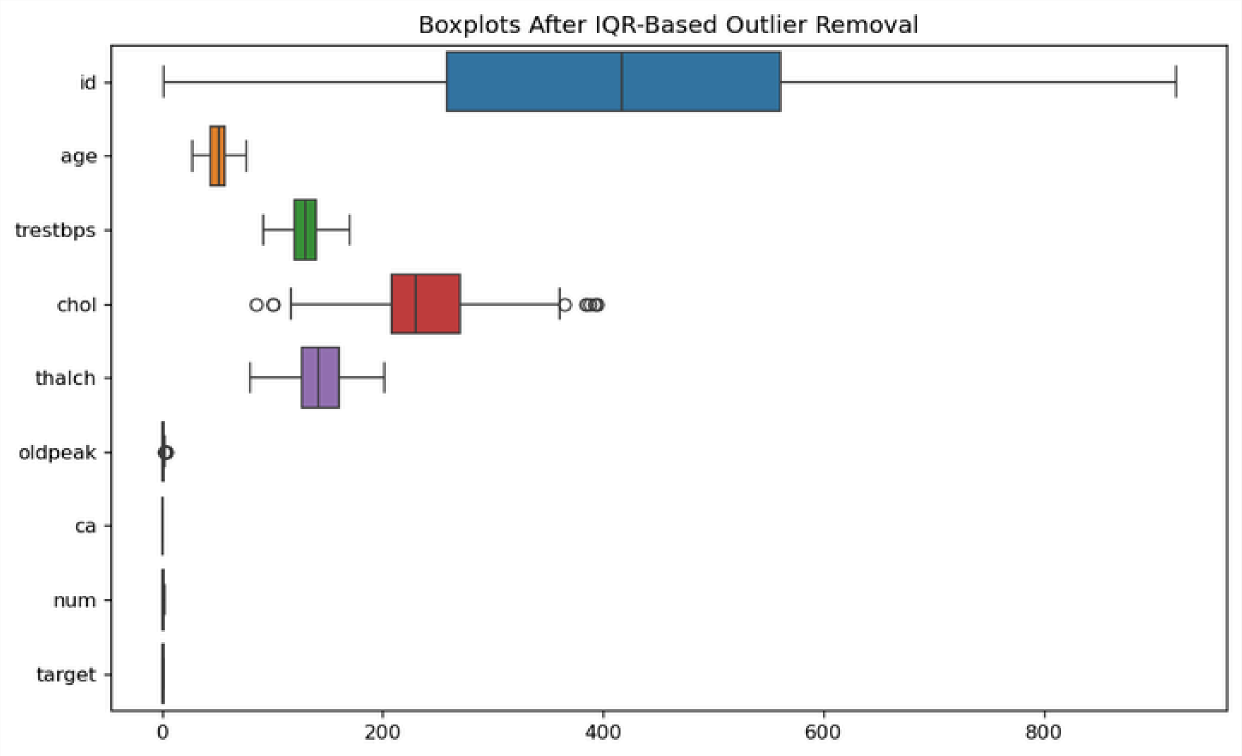
4.Data Preprocessing

- Converted the multi-class target variable into a binary classification variable to simplify the prediction task.

  0 → No Heart Disease

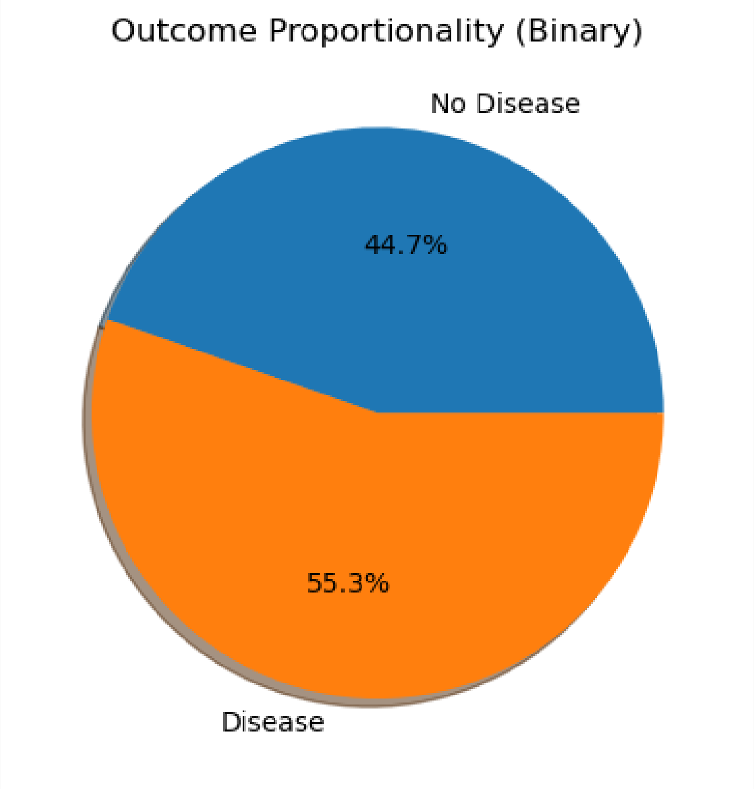
  1 → Presence of Heart Disease

- Handled missing values in numerical features using median imputation for numerical features to ensure data completeness and robustness against outliers.
- Handled missing values in categorical features using most frequent value imputation to maintain data consistency.
- Applied the Interquartile Range (IQR) method to detect and remove outliers from numerical features to improve data quality.

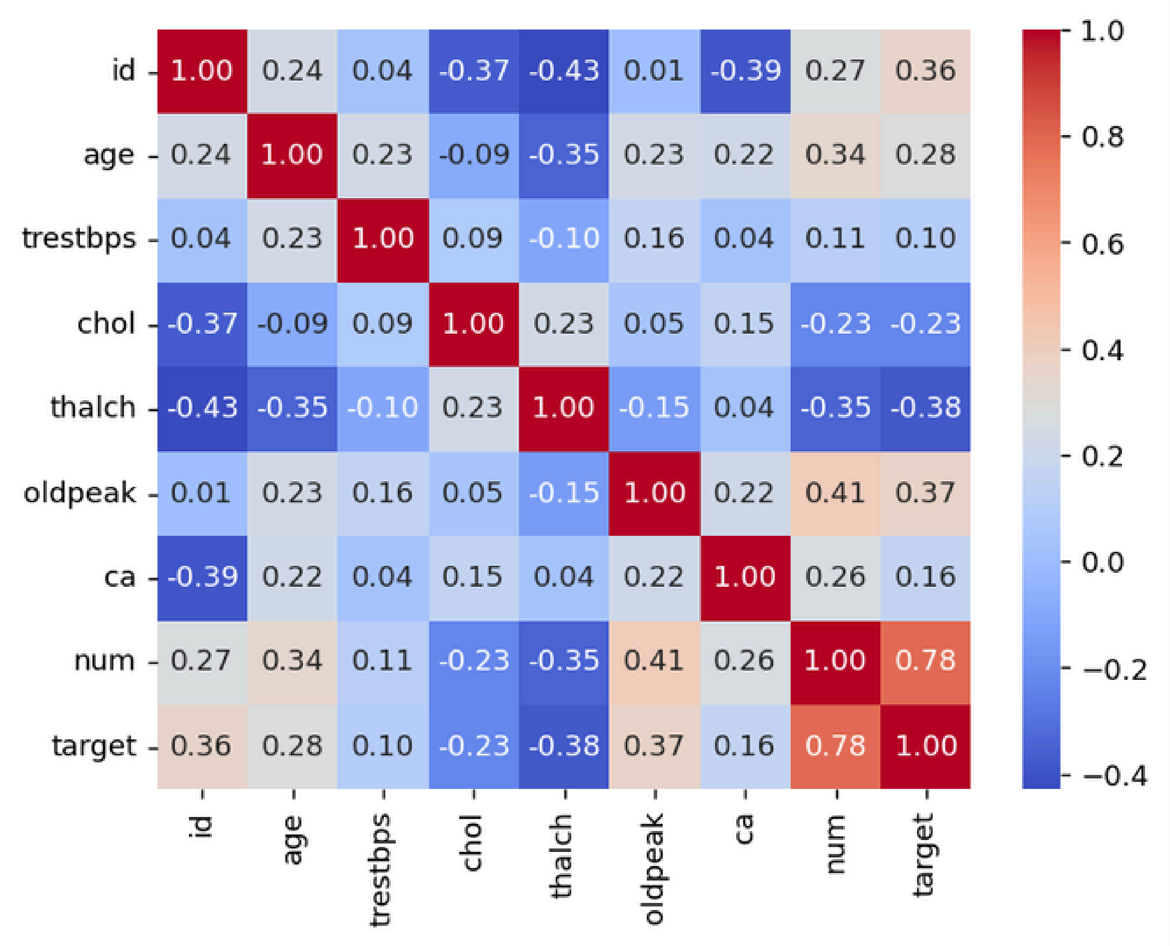


5.Post-Cleaning EDA

- Visualized the distribution of the binary target classes using a pie chart to assess class proportion after preprocessing.



- Computed the correlation matrix of numerical features to analyze relationships between variables after preprocessing.



6.Train–Test Split

- Divided the dataset into training and testing sets with an 80:20 ratio to evaluate model performance on unseen data.

7.Feature Scaling

- Normalization (Min-Max Scaling) rescales feature values to a fixed range, typically between 0 and 1.
- Standardization (StandardScaler) transforms features to have zero mean and unit standard deviation.

8.Handle Class Imbalance Using SMOTE

- Addressed class imbalance using the Synthetic Minority Oversampling Technique (SMOTE) to improve the model's ability to learn minority class patterns.
- SMOTE was applied only to the training dataset to generate synthetic samples of the minority class while preserving the integrity of the test data.

9.Model Development and Training

SVM (Support Vector Machines)

- SVM constructs an optimal decision boundary that maximizes the margin between disease and non-disease classes.
- Multiple kernel functions (linear, radial basis function (RBF), polynomial, and sigmoid) were evaluated to determine the most suitable decision boundary for the data.

Random Forest

- Implemented a Random Forest classifier, an ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting.
- Applied RandomizedSearchCV to perform hyperparameter optimization by exploring multiple parameter combinations such as number of trees, tree depth, and minimum samples for node splitting.

XGBoost

- Implemented XGBoost (Extreme Gradient Boosting), an advanced ensemble learning algorithm that builds sequential decision trees to improve prediction accuracy.
- Applied RandomizedSearchCV to perform hyperparameter tuning, optimizing parameters such as number of estimators, learning rate, tree depth, and sampling ratios.

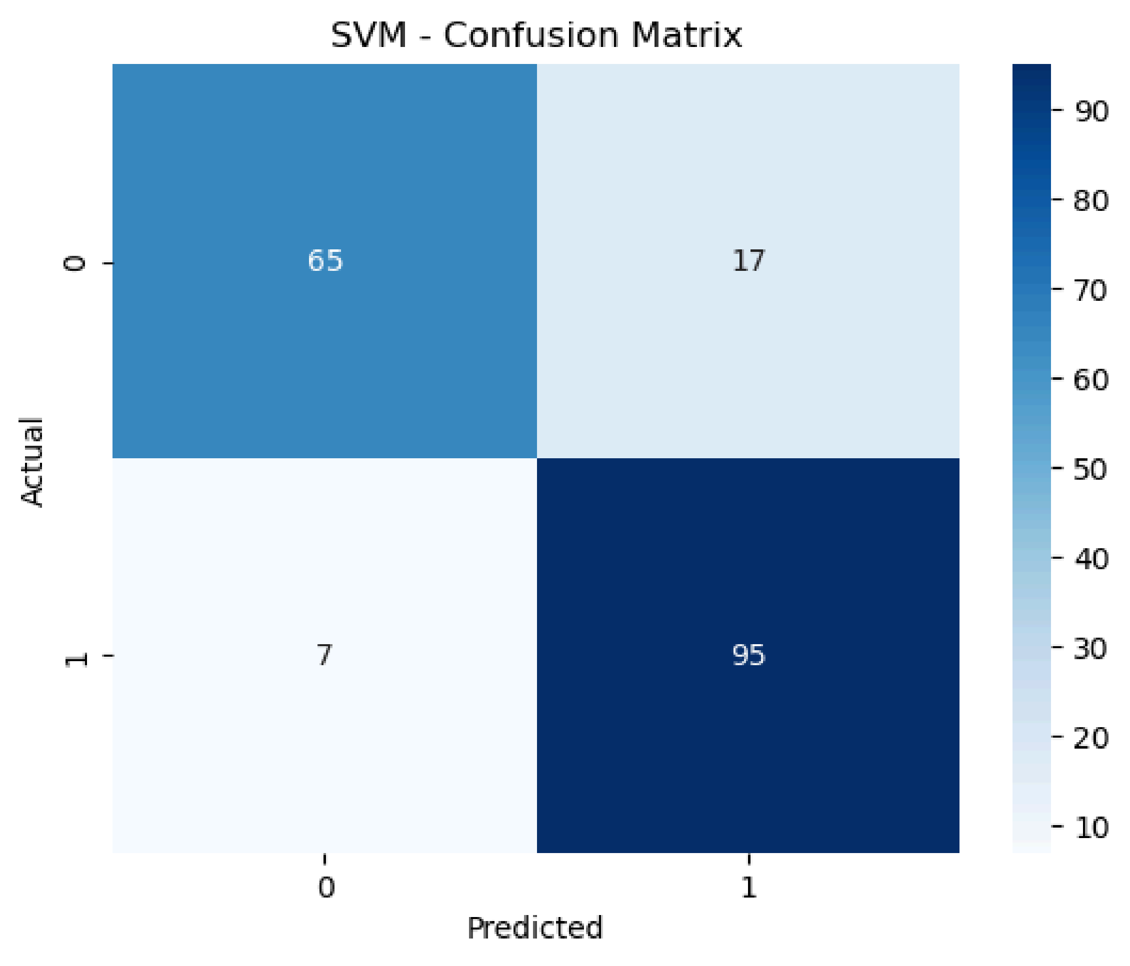
LightGBM (Light Gradient Boosting Machine)

- Gradient boosting framework designed for efficient training and high predictive performance on structured datasets.
- Key parameters such as number of estimators, learning rate, and tree depth were configured to control model complexity and prevent overfitting.

10.Model Evaluation

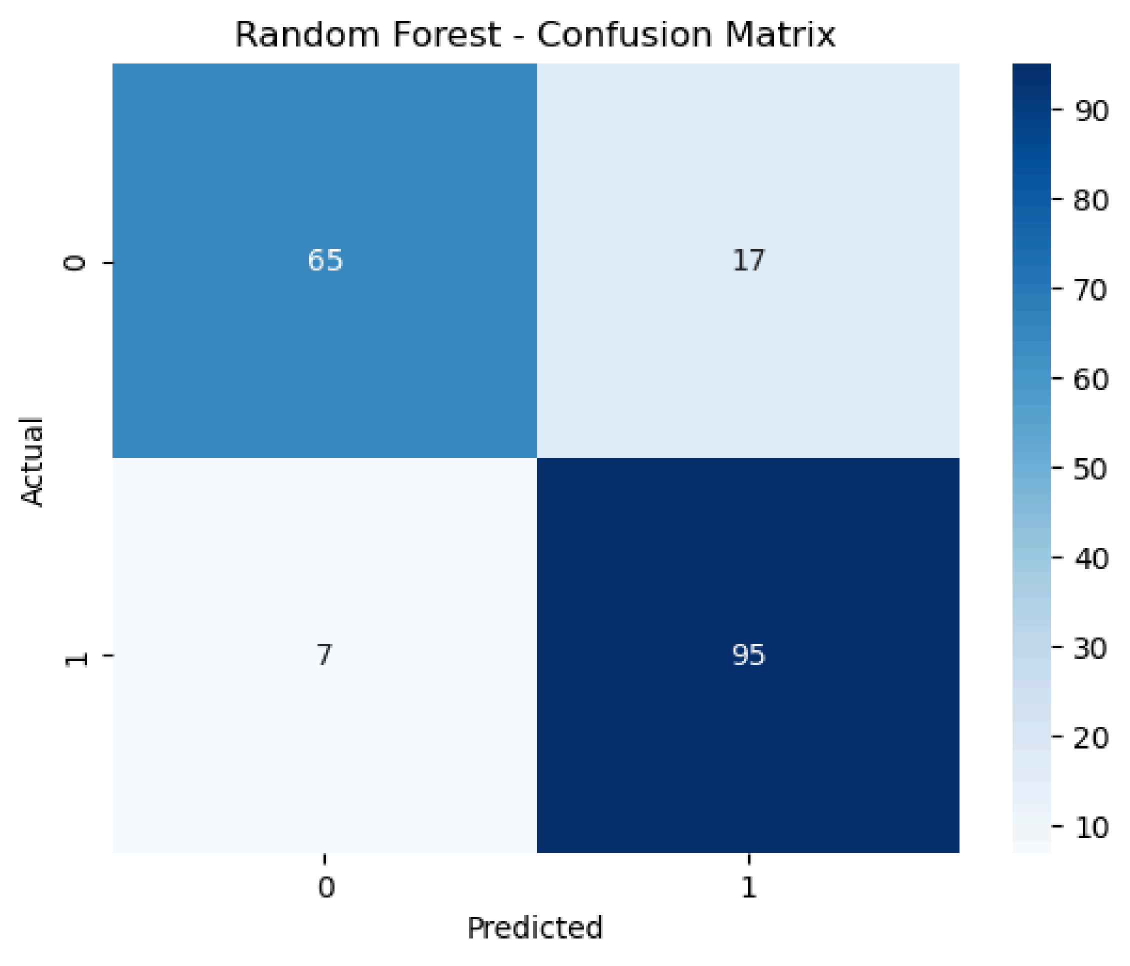
SVM

Accuracy: 0.8695652173913043



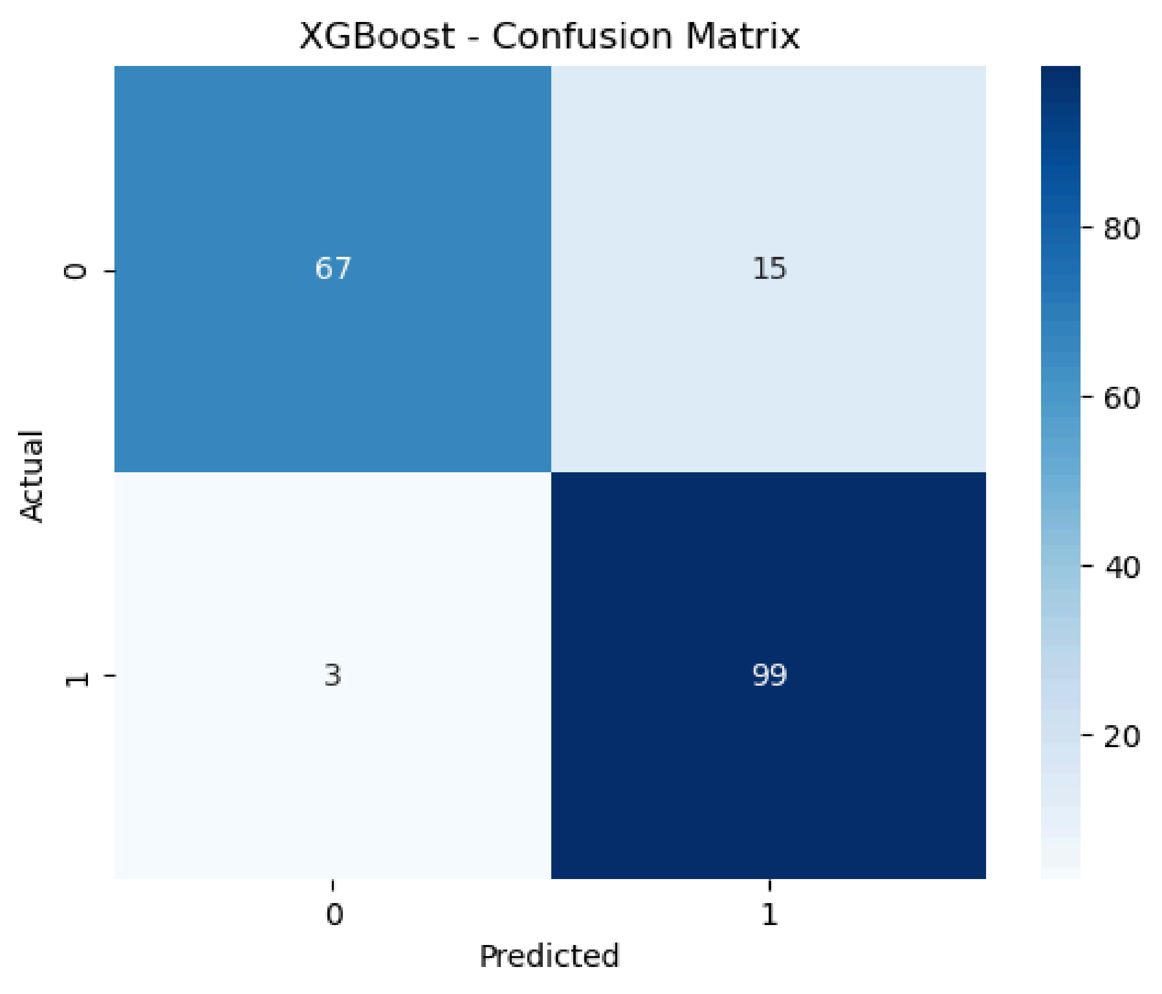
Random Forest

Accuracy:0.8695652173913043



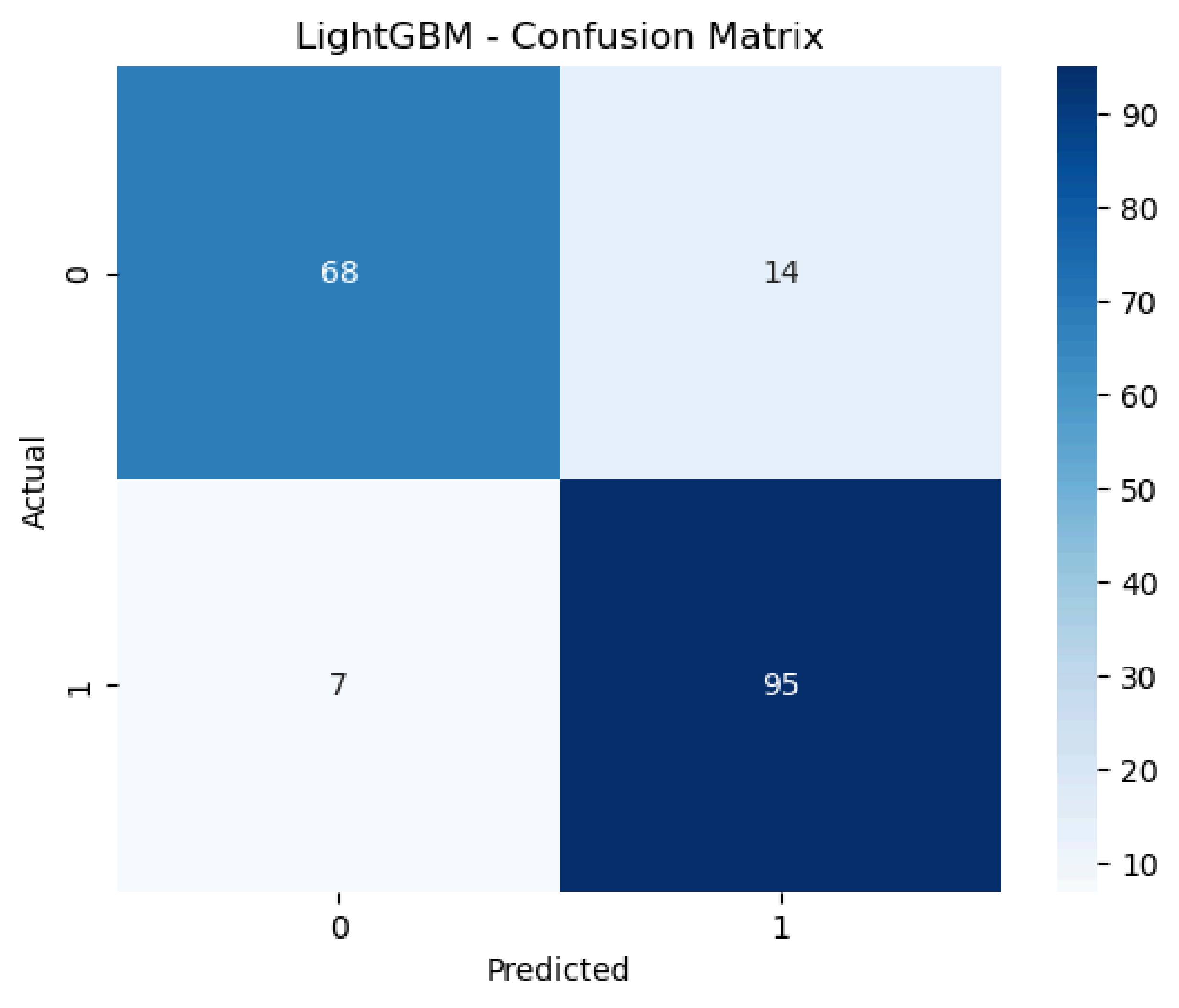
XGBoost

Accuracy: 0.9021739130434783



LightGBM

Accuracy: 0.8858695652173914

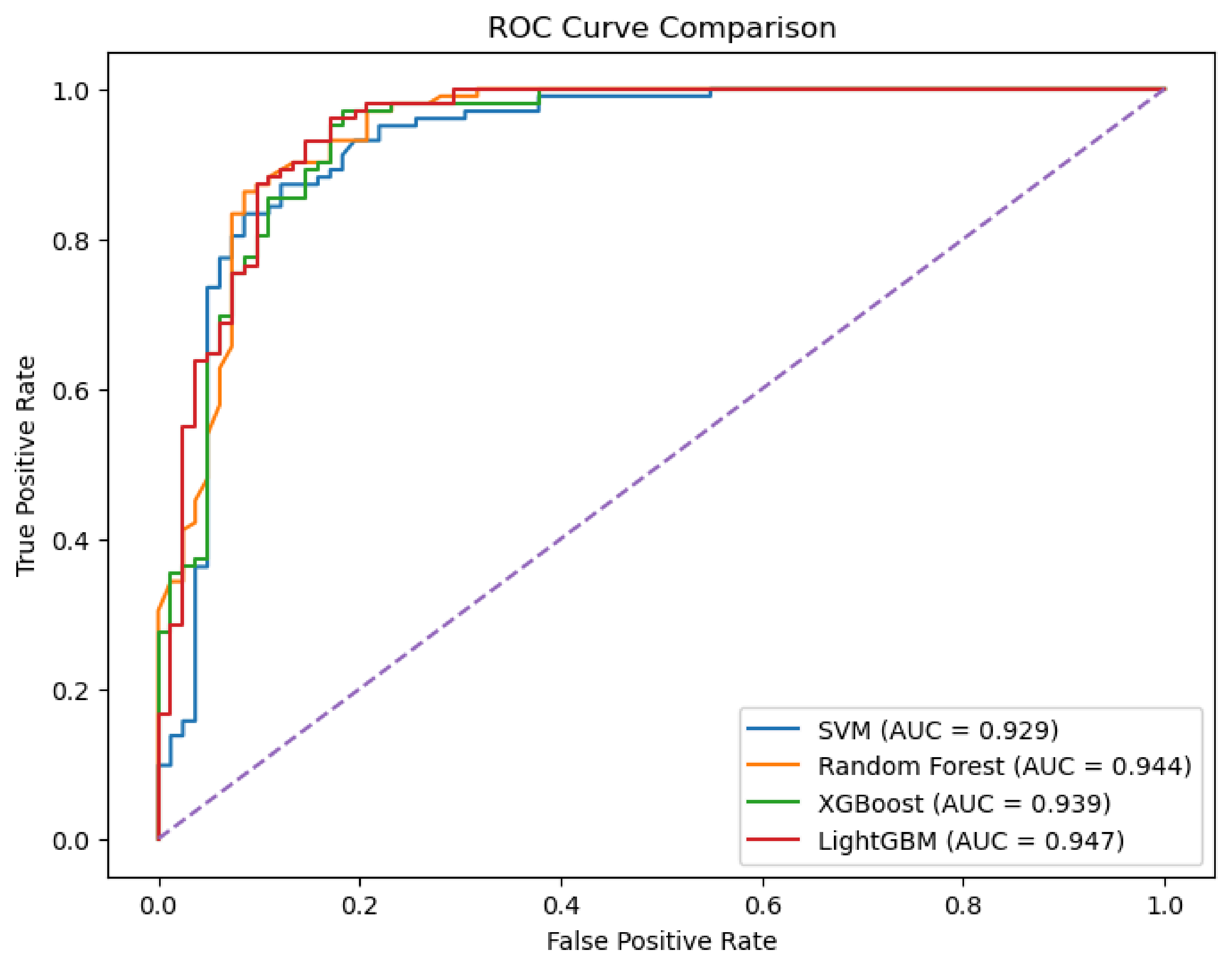


11.Performance Comparison

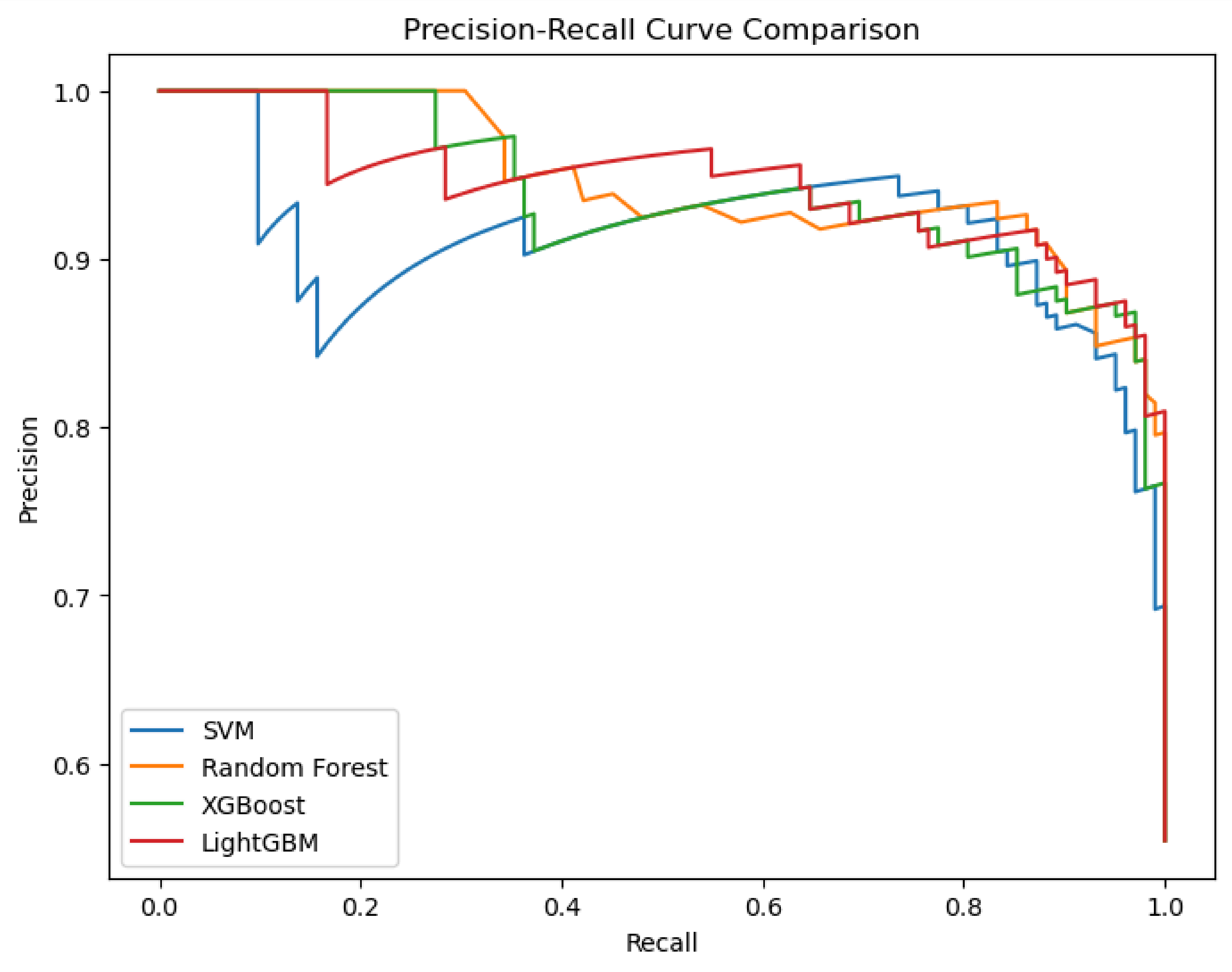
- comparative performance of the implemented machine learning models based on Accuracy, Precision, Recall, and F1 Score.

	Accuracy	Precision	Recall	F1 Score
XGBoost	0.902174	0.868421	0.970588	0.916667
LightGBM	0.885870	0.871560	0.931373	0.900474
SVM	0.869565	0.848214	0.931373	0.887850
Random Forest	0.869565	0.848214	0.931373	0.887850

- The Receiver Operating Characteristic (ROC) curve is used to evaluate the ability of classification models to distinguish between positive and negative classes across different decision thresholds.
- The Area Under the Curve (AUC) provides a single measure of overall model performance, where higher values indicate better classification capability.



- The Precision-Recall (PR) curve evaluates the performance of classification models by analyzing the trade-off between precision and recall at different decision thresholds.



Conclusion

- In this study, a machine learning–based framework was developed to predict the presence of heart disease using clinical attributes.
- Four classification models were implemented and evaluated using standard performance metrics.
- The experimental results demonstrate that ensemble learning methods provide effective and reliable performance for heart disease prediction.
- Among the evaluated models, XGBoost achieved the highest overall performance, demonstrating strong predictive capability with balanced precision and recall.

Thank you